

Direct kinematics

Fundamentals of Robotics

University of Trento

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Outline

Definitions on robotic manipulators

Direct Kinematics

The Denavit-Hartenberg Convention

Examples

- Three-link Planar Arm

- Spherical Arm

- Antropomorphic Arm

- Spherical Wrist

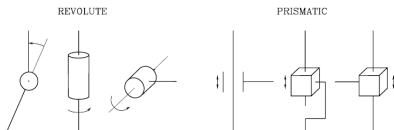
Joint Space and Operational Space and Work Space

Useful definitions

Manipulators

Let us recall a few definitions related to robotic manipulators

- ▶ A manipulator consists of a series of rigid bodies said *links*
- ▶ Links are connected by kinematic pairs said *joints*
- ▶ Joints are of two types
 - ▶ Revolute
 - ▶ Prismatic
- ▶ the two types of joints are graphically represented like this



Manipulators

- ▶ The whole structure forms a kinematic chain
- ▶ One end of the chain is connected to a base
- ▶ the other end is connected to an end-effector (gripper, tool) that is used for the manipulation activities
- ▶ A kinematic chain is characterised by a number of degrees of freedom (DoF) that uniquely determine its *posture*
- ▶ The posture is given by the pose of all the links
- ▶ Each DoF is associated with a joint variable.
- ▶ The goal of *direct kinematics* is to find the pose of the end effector given the joint variables

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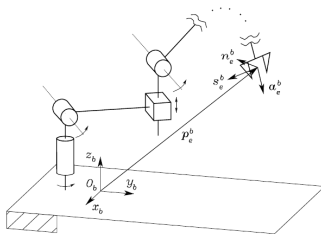
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Joint Space and Operational Space and Work Space

Useful definitions

Pose of the end-effector



Given a fixed frame $O_b - x_b y_b z_b$ attached to the base and a frame attached to the end-effector, the pose of the latter is given by the homogeneous transformation matrix

$$T_e^b(q) = \begin{bmatrix} R_e^b(q) & p_e^b(q) \\ 0^T & 1 \end{bmatrix} = \begin{bmatrix} n_e^b(q) & s_e^b(q) & a_e^b(q) & p_e^b(q) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

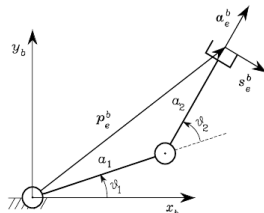
where:

- ▶ q is the n -vector containing the joint variables
- ▶ n_e, s_e, a_e are the unit vectors associated with the end effect and are chosen accounting for its geometry.
- ▶ If the end-effector is a two fingered gripper, a_e is chosen in the approach direction, s_e is chosen in the sliding direction and n_e is chosen orthogonal to the other two so that the frame (n_e, s_e, a_e) is right-handed.

All the elements of the matrix are functions of the joint variables q .

Example

In the easiest cases it is possible to compute the direct kinematics by straight-forward geometric considerations. Consider the example of a two link planar manipulator



With some simple maths you come up with

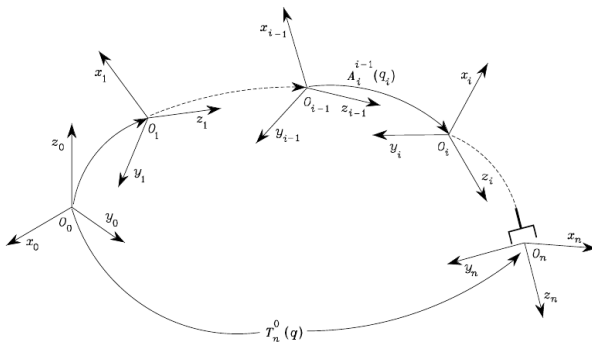
$$T_e^b(q) = \begin{bmatrix} n_e^b & s_e^b & a_e^b & p_e^b \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & s_{12} & c_{12} & a_1 c_1 + a_2 c_{12} \\ 0 & -c_{12} & s_{12} & a_1 s_1 + a_2 s_{12} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

($s_i = \sin(\theta_i)$, $c_i = \cos(\theta_i)$, $s_{12} = \sin(\theta_1 + \theta_2)$, $c_{12} = \cos(\theta_1 + \theta_2)$).

In more complex case the direct derivation can be much more difficult.

Let us look for a general procedure.

Open chains



- ▶ Let us consider an open chain with $n + 1$ links connected by n joints (no loops between the links).
- ▶ Link 0 is fixed to the ground
- ▶ If we attach a frame to each link the pose of the frame $i + 1$ will be determined from the pose of frame i through a simple transformation.
- ▶ The idea is that we can construct recursively the transformation between the 0-th and the n -th link recursively

$$T_n^0(q) = A_1^0(q_1) A_2^1(q_2) \dots A_n^{n-1}(q_n)$$

where $A_i^{i-1}(q_i)$ is an homogeneous transformation matrix and is a function of the joint variable q_i

- ▶ To complete the picture, we need an additional transformation from the base to the first link and one from the last link to the end effector (which are typically constant)

$$T_e^b(q) = T_0^b T_n^o(q) T_e^n.$$

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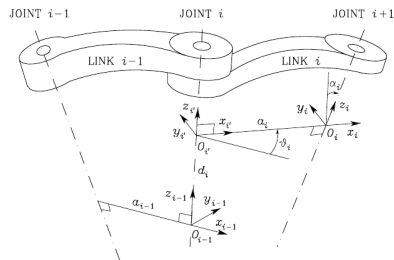
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Useful definitions

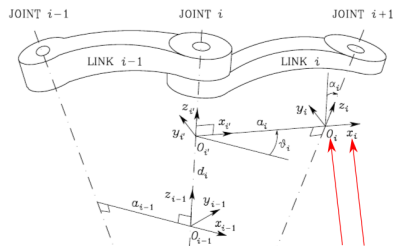
Denavit-Hartenberg Convention

- ▶ In order to find the direct kinematics of an open chain along the lines discussed before we need an effective way to place the different frames
- ▶ The D-H method goes exactly in this direction
- ▶ Suppose that axis i connects link i with joint $i + 1$
- ▶ Axis z_i is chosen aligned to the Axis i
- ▶ We still have to choose the origin O_i and the axes x_i and y_i



Denavit-Hartenberg Convention - 1

- ▶ Define the *common normal* to two lines as the line containing the minimum distance segment between two lines
 1. Uniquely defined if the lines are not co-planar
 2. Infinite number of common normals if the lines are co-planar and parallel
 3. One point (degenerate situation) if the lines are planar and intersect
- ▶ The origin O_i is chosen at the intersection between the common normal and the axis z_i
- ▶ Axis x_i is chosen along the common normal in the direction from axis z_{i-1} to axis z_i
- ▶ Axis y_i is chosen so that the frame is right-handed



Denavit-Hartenberg Convention - 1

Indeterminate Conditions

- ▶ For Frame 0 only z_0 is specified. Then O_0 and x_0 can be arbitrary.
- ▶ For Frame n there is no joint $n + 1$. So z_n is not uniquely defined. Axis x_n remains normal to axis z_{n-1} . Being joint n typically revolutes, z_n is aligned with z_{n-1}
- ▶ When two axes are parallel, there is no unique common normal
- ▶ When two axes intersect there is not a unique definition for axis x_i
- ▶ When joint i is prismatic, the direction of z_{i-1} is arbitrary

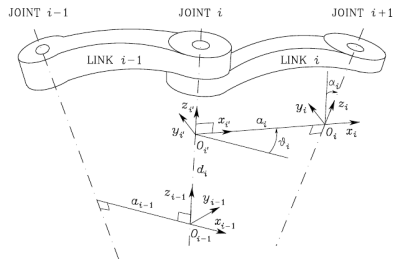
In all these conditions we try to make a choice in order to simplify the procedure (i.e. choosing parallel axes for consecutive frames).

Denavit-Hartenberg Convention - 2

Let $O_{i'} - x_{i'}y_{i'}z_{i'}$ be a frame parallel to $O_{i-1} - x_{i-1}y_{i-1}z_{i-1}$ with the origin on the intersection between the common normal and z_{i-1} .

Introduce the following parameters.

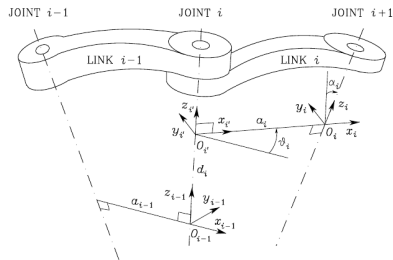
- ▶ a_i distance between O_i and $O_{i'}$
- ▶ d_i coordinate of $O_{i'}$ along z_{i-1}
- ▶ α_i rotation angle between z_{i-1} and z_i about x_i (positive if counter-clock wise)
- ▶ θ_i rotation angle between x_{i-1} and x_i about z_{i-1} (positive if counter-clock wise)



Denavit-Hartenberg Convention - 3

Fixed parameters a_i and θ_i . Related to the geometry of the links.

Variable parameters: θ_i for revolute joints, d_i for prismatic joints.



Denavit-Hartenberg Convention - 4

Construction of the transformation matrix between link $i-1$ and link i .

- ▶ Start from a frame aligned with Frame $i-1$
- ▶ Translate the frame by d_i along axis z_{i-1} and rotate it by θ_i about axis z_{i-1} aligning it with frame i' :

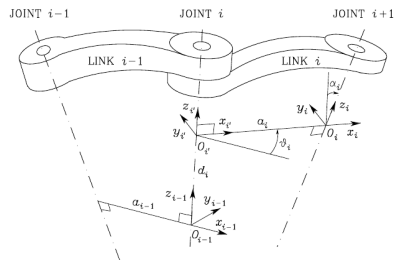
$$A_{i'}^{i-1} = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ▶ Translate the frame aligned with Frame i' by a_i along axis $x_{i'}$ and rotate it by α_i about axis $x_{i'}$ aligning it with Frame i :

$$A_i^{i'} = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ▶ The complete transformation is then given by:

$$A_i^{i-1}(q_i) = A_{i'}^{i-1} A_i^{i'} = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Summary of the procedure

1. Find the joint axes and assign $z_0, \dots, z_{n-1}$
2. Complete Frame 0 by choosing x_0, y_0 so that the frame is right-handed (useful to have it coincident with the frame's basis)
3. For all $i \in \{1, \dots, n-1\}$
 - 3.1 locate the origin O_i at the intersection between z_i and z_{i-1} . If axes z_{i-1} and z_i are parallel then:
 1. if joint i is revolute locate O_i so that $d_i = 0$, 2. if joint i is prismatic locate O_i at a meaningful position for the range (e.g., the mechanical limit).
 - 3.2 Choose axis x_i along the common normal between axes z_{i-1} and z_i , pointing from Joint i to Joint $i+1$
 - 3.3 Choose axis y_i so that the frame is right handed ($y_i = z_i \times x_i$).
4. Choose Frame n . If Joint n is revolute then align z_n with z_{n-1} , if it is prismatic z_n is arbitrary. For the other axes proceed as for the other joints.
5. For all $i \in \{1, \dots, n\}$ for a table with $a_i, d_i, \alpha_i, \theta_i$
6. Compute the homogeneous transformation $A_i^{i-1}(q_i)$ and then form the homogeneous transformation

$$T_n^0(q) = A_1^0 \dots A_n^{n-1}.$$

7. Given T_0^b and T_e^n complete the direct kinematic function:

$$T_e^b(q) = T_0^b T_n^0 T_e^n.$$

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Three Link Planar Arm

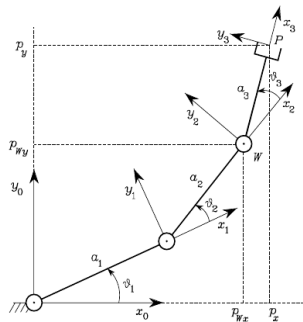
- ▶ All joints are revolute and have parallel axes.
- ▶ The direction of x_0 is arbitrary. For x_1 , x_2 , and x_3 the most convenient choice is to have them aligned with the links.
- ▶ The d_i parameters are null and the joint variables are essentially the angles between the links.
- ▶ The D-H table is the following.

Link	a_i	α_i	d_i	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3

- ▶ For all joints we have a rotation about the z_{i-1} axis and a translation of the link length along the direction of the link. Hence

$$A_i^{i-1}(\theta_i) = \begin{bmatrix} c_i & -s_i & 0 & a_i c_i \\ s_i & c_i & 0 & a_i s_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

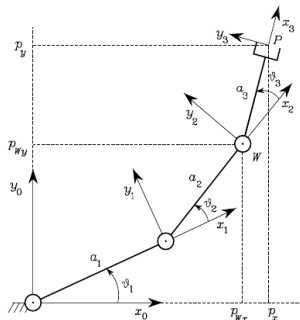
$$T_3^0(q) = A_1^0 A_2^1 A_3^2 = \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ s_{123} & c_{123} & 0 & a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Three Link Planar Arm - 1

- Normally, the frame attached to the end effector has z_e as approach angle. So we still need to rotate Frame 3 by 90 degrees about the y_3 axis:

$$T_e^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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Spherical Arm

- ▶ The origin O_0 is at the intersection of axis z_0 with axis z_1 (hence $d_1 = 0$). Likewise, The origin of Frame 2 is at the intersection between z_1 and z_2 .

- ▶ The table of the DH parameters is:

Link	a_i	α_i	d_i	θ_i
1	0	$-\frac{\pi}{2}$	0	θ_1
2	0	$\frac{\pi}{2}$	d_2	θ_2
3	0	0	d_3	0

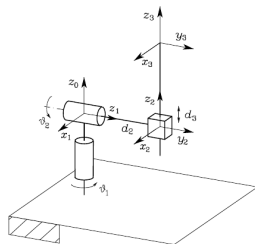
- ▶ The transformation matrices are the following:

Rotate θ_1 about z_0 and $-\frac{\pi}{2}$ about x'_0

$$A_1^0(\theta_1) = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translate d_2 along z_1 , rotate θ_2 about z_1 , and then $\frac{\pi}{2}$ about x'_1 :

$$A_2^1(\theta_2) = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Spherical Arm - 1

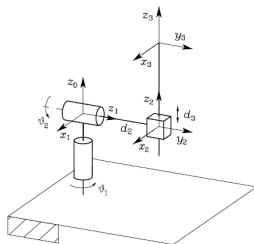
- The Final transformation is a mere translation along z_2

$$A_3^2(d_3) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which leads to

$$T_3^0(q) = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & c_1 s_2 d_3 - s_1 d_2 \\ s_1 c_2 & c_1 & s_1 s_2 & s_1 s_2 d_3 + c_1 d_2 \\ -s_2 & 0 & c_2 & c_2 d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d_3 \end{bmatrix}$$



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Antropomorphic Arm

The arm is a two-link plan arm with an additional rotation about an x-axis.

- ▶ Axes z_0 and z_1 intersect, while z_1 and z_2 are parallel.

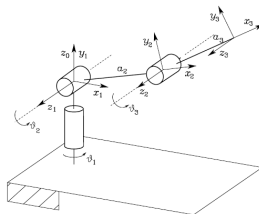
- ▶ The DH parameters are as in the following:

Link	a_i	α_i	d_i	θ_i
1	0	$\frac{\pi}{2}$	0	θ_1
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3

- ▶ The homogeneous transformation are actually a combination of the one seen for the spherical arm and the one for the planar arm.

$$A_1^0(\theta_1) = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

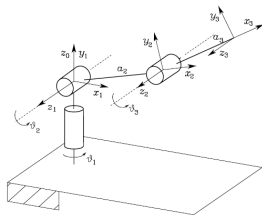
$$A_i^{i-1}(\theta_i) = \begin{bmatrix} c_i & -s_i & 0 & a_i c_i \\ s_i & c_i & 0 & a_i s_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \text{ for } i = 2, 3$$



Anthropomorphic Arm - 1

The total transformation is given by

$$T_3^0(q) = A_1^0 A_2^1 A_3^2 = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1(a_2 c_2 + a_3 c_{23}) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1(a_2 c_2 + a_3 c_{23}) \\ s_{23} & c_{23} & 0 & a_2 s_2 + a_3 s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$



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Spherical Wrist

- For this we start the numbering from 3 because it is a wrist (mounted on the tip of a manipulator)
- Once z_3 , z_4 , z_5 and x_3 are fixed, x_4 and x_5 are determined arbitrarily.
- The DH parameters are in the following.

Link	a_i	α_i	d_i	θ_i
4	0	$-\frac{\pi}{2}$	0	θ_4
5	0	$\frac{\pi}{2}$	0	θ_5
6	0	0	d_6	θ_6

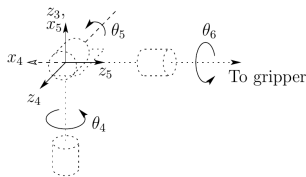
- The transformation matrices are the following:

Rotate θ_4 about z_3 and $-\frac{\pi}{2}$ about x_3' :

$$A_4^3(\theta_4) = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotate θ_5 about z_4 and $-\frac{\pi}{2}$ about x_4' :

$$A_5^4(\theta_5) = \begin{bmatrix} c_5 & 0 & -s_5 & 0 \\ s_5 & 0 & c_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Spherical Wrist - 1

- The transformation matrices for the last link:

Translate of d_6 about z_5

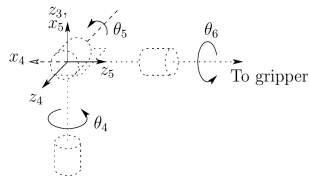
Rotate θ_6 about z_5 and $-\frac{\pi}{2}$ about x_5' :

$$A_5^6(\theta_6) = \begin{bmatrix} c_6 & 0 & -s_6 & 0 \\ s_6 & 0 & c_6 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final Result :

$$T_6^4(q) = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 & c_4 s_5 d_6 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 & s_4 s_5 d_6 \\ -s_5 c_6 & s_5 s_6 & c_5 & c_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$q = \begin{bmatrix} \theta_4 \\ \theta_5 \\ \theta_6 \end{bmatrix}$$



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Operational space

- ▶ The operations of a manipulator are typically specified in terms of position and orientation of the end-effector
- ▶ This requires specifying three coordinates for the position p_e and the three vectors defining the frame attached to the end effector (n_e, s_e, a_e)
- ▶ The latter is quite complicated because providing nine elements is redundant because they are bound to be orthonormal.
- ▶ It is much better to adopt a non redundant way notation (e.g., by using Euler's angles):

$$x_e = \begin{bmatrix} p_e \\ \phi_e \end{bmatrix}$$

- ▶ The vector x_e is specified in the space where the end-effector operates (operational space).

Joint space

- ▶ The joint space (or configuration space) denote the space in which the vector

$$\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$$

is defined, where

$$q_i = \begin{cases} \theta_i & \text{revolute joint} \\ d_i & \text{prismatic joints.} \end{cases}$$

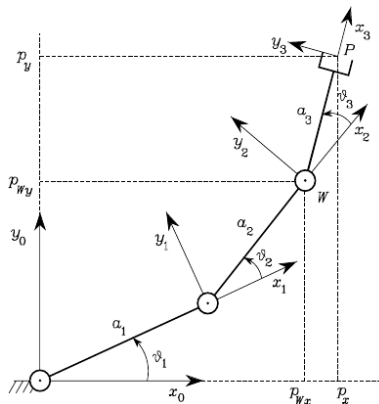
- ▶ The direct kinematics equations are defined by a function

$$\mathbf{x}_e = \mathbf{k}(\mathbf{q}).$$

Example

It is easy to see that the direct equation are in this case defined by:

$$\begin{aligned} x_e &= \begin{bmatrix} p_x \\ p_y \\ \phi \end{bmatrix} = K(q) = \\ &= \begin{bmatrix} a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ \phi_1 + \phi_2 + \phi_3 \end{bmatrix} \end{aligned}$$



Workspace

- ▶ The workspace is a performance index for the robot
- ▶ It is the region described by the origin of the frame attached to the end-effector for all possible motions at the joints
- ▶ We distinguish between
 - ▶ *Reachable space*: points that can be reached by the end-effector with at least one configuration
 - ▶ *Dexterous space*: points that can be reached with more than one configuration
 - ▶ Clearly the Dexterous Space is a subspace of the reachable space

Reachable Workspace

- ▶ The reachable workspace can be computed by simply considering the evolution of

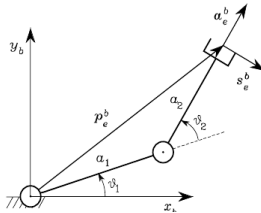
$$p_e = p_e(q), \quad q_{im} \leq q_i \leq q_{iM}, \forall i \in \{1, \dots, n\}.$$

where each joint variable is made to vary within its minimum q_{im} and maximum q_{iM} boundaries.

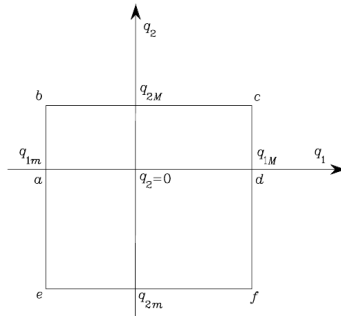
- ▶ The volume described in this way is finite, closed, and connected (for the continuity of $p_e(q)$); thus its frontier (the bordering surface) becomes very important.

Example

Consider the two link planar manipulator

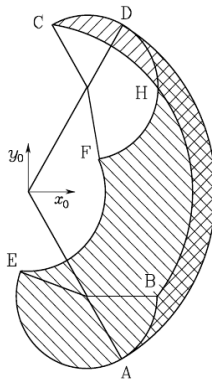
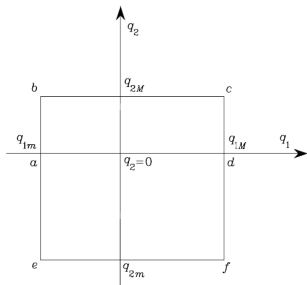


The two link variables take values in a rectangular region



Example (continued)

The best way to construct the workspace is to make the arm move following the perimeter of the region of ammissible configurations: Segments ab , bc , cd , da , ae , ef , fd . We show the trajectory followed by the end-effector.



Outline

Definitions on robotic manipulators

Direct Kinematics

The Denavit-Hartenberg Convention

Examples

- Three-link Planar Arm

- Spherical Arm

- Antropomorphic Arm

- Spherical Wrist

Joint Space and Operational Space and Work Space

Useful definitions

Accuracy

- ▶ In the expression of the direct kinematics

$$\mathbf{x}_e = \mathbf{k}(\mathbf{q}),$$

we have not explicitly shown the role of the DH parameters (e.g., the lengths of the links)

- ▶ All these mechanical parameters are known with some degree of inaccuracy (mechanical tolerance).
- ▶ Therefore when we set a value for the joint parameters $\mathbf{q}$, we will have some possible deviation in $\Delta\mathbf{x}_e$ in the position of the end-effector
- ▶ This deviation is called accuracy and is usually below 1mm.
- ▶ The accuracy depends on the geometry of the manipulator (what is the maximum reach) and on the position in the workspace

Repeatability

- ▶ Repeatability is another important parameter and measures the ability of the manipulator to return to a previously visited position.
- ▶ This parameter is very important for teach-by-showing examples and it depends also on the precision of transducers and controllers
- ▶ For a manipulator with a maximum reach of 1.5m, accuracy is usually in the range $[0.2, 1]$ mm while repeatability is in the range $[0.02, 0.2]$ mm.

Kinematic Redundancy

- ▶ A manipulator is kinematically redundant if its number of DoFs is greater than the number of variables needed to describe a task.
- ▶ From what we have seen, there is an intrinsic redundancy when the dimension n of the joint space is larger than the dimension m of the operational space.
- ▶ In general if the task requires a specification of r dimension ($r \leq m$) we have redundancy of $r < n$.
- ▶ As an example, the 3 DoF planar manipulator ($n = 3$) is redundant if we only impose the position of the end effector $r = 2$, it is not if we impose also the orientation $r = 3$
- ▶ Redundancy is related to the task rather than to the system.
- ▶ Often a redundant manipulator is useful to increase the dexterity or the mobility (e.g., the human arm has 7 DoF)